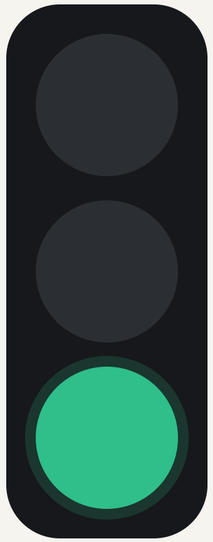
Signal Learner

Stevens Creek congestion led us to build a working simulation that tests timing changes for an engineer.



MONTA VISTA AI CLUB
MONTA VISTA HIGH SCHOOL, CUPERTINO

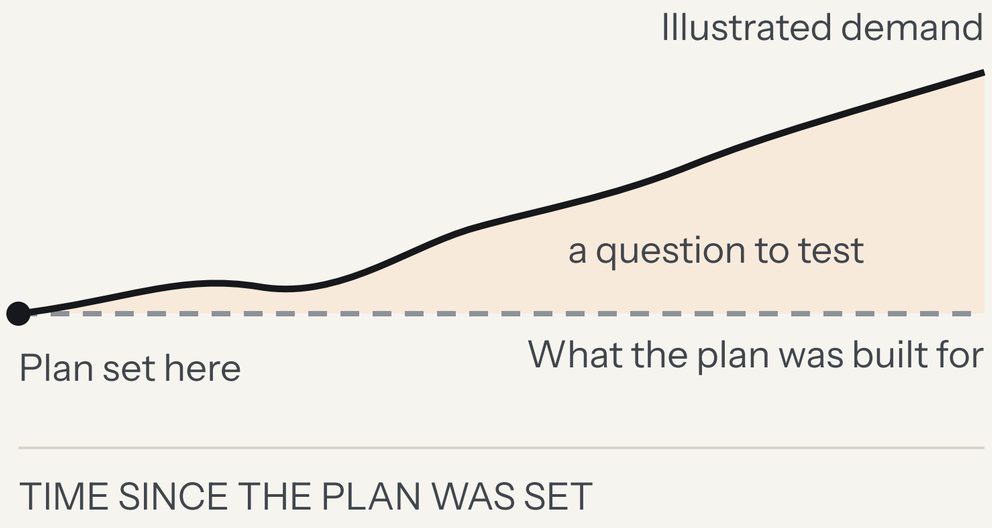
THE PROBLEM

Stevens Creek backs up. Which timing changes are worth testing?

“the traffic was backed up all the way to Glenbrook today.”

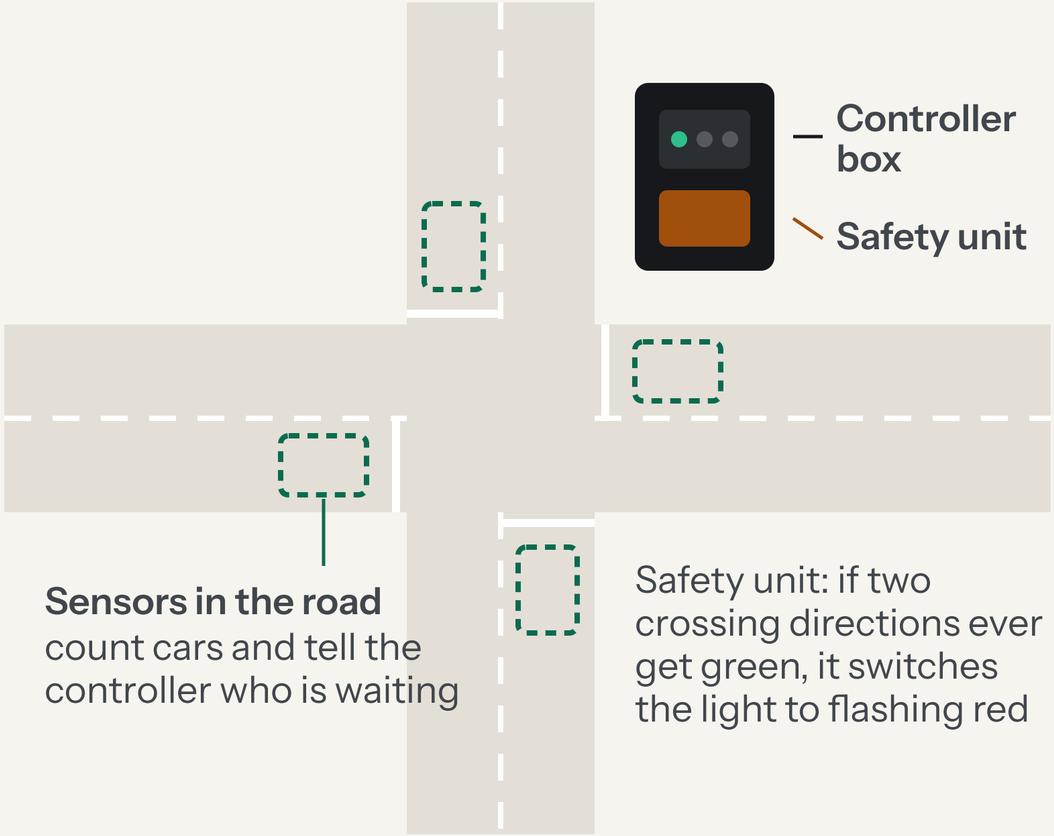
TEAM OBSERVATION, OCT 21

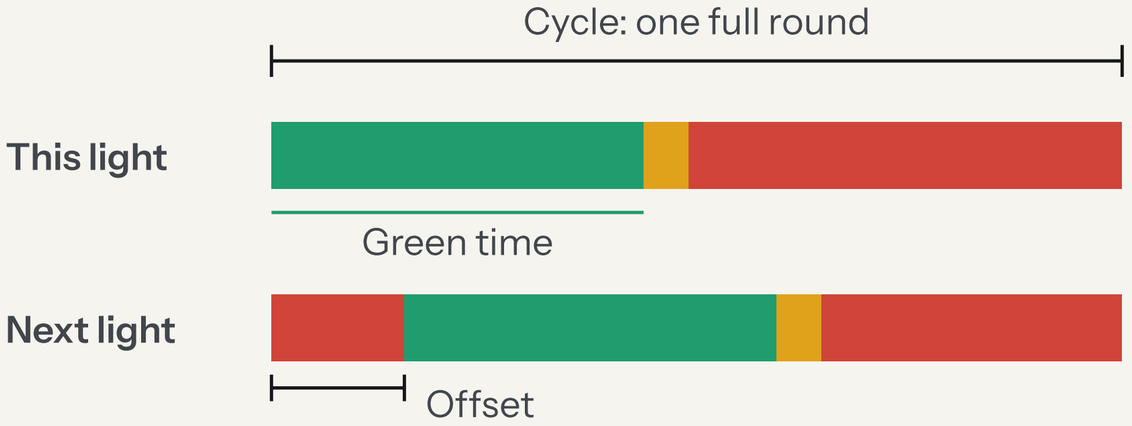
Our observation, not a measured congestion study. In a Nov 1 email, the city said Stevens Creek uses time-of-day timing plans. We used VTA 2024 counts to start a simulation study.



Sketch of the idea, not measured data.

A small computer at the corner follows a timing plan.





Cycle One full round of greens for every direction.

Green time How long one direction gets to go.

Offset How many seconds after its neighbor a light turns green. Neighbors share one cycle, so cars can hit greens in a row.

IS THIS ALREADY SOLVED?

Existing systems matter. Our contribution is a workflow you can inspect and test.

CHANGE THE TIMING ON THE GO

SCOOT SCATS InSync Surtrac SynchroGreen

RECOMMENDS, ENGINEERS DECIDE

Google Green Light

> Signal systems already adapt timing or recommend changes for engineers.
>
> **Our contribution: trace observations to a model, a tested candidate and a recommendation.**
>
> A direct comparison with Google Green Light or an engineer's plan remains future work.

WHAT A REVIEWER CAN INSPECT

Observe. Test. Recommend.

The inputs, search rules, recommendation card and held-out results are saved.

OUR IDEA

Six steps, and it never touches a real light.

1 Map

Draw the road from the public map. Old timing keeps running.

2 Watch

Read the road sensors for 3 simulated PM rush hours. Change nothing.

3 Learn

Learn the traffic pattern and build a practice copy of the road.

4 Practice

Try many timings on the copy, over many rush hours.

5 Recommend

Suggest one bounded change and its reason. An engineer approves or rejects.

6 Check

Monitor simulated observations and apply the tested rollback rule.

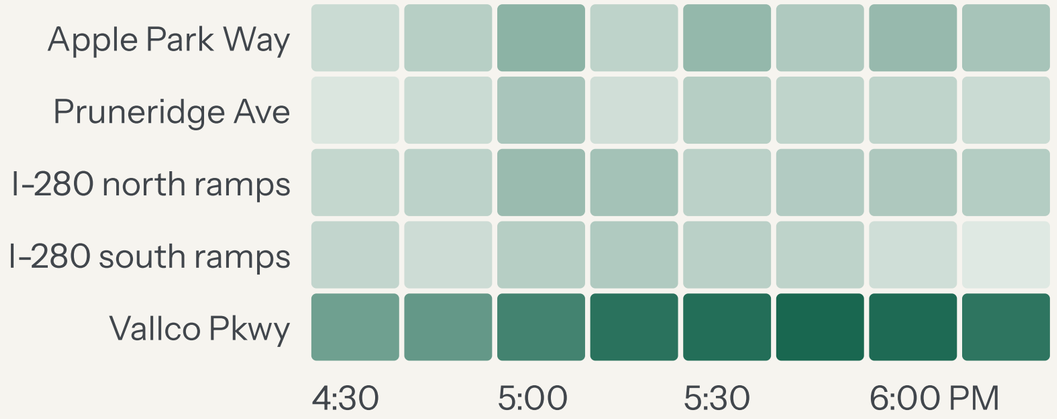
It never touches a real light. In our tests, the simulator applies an approved change.

WHERE THE AI IS

SIMULATED

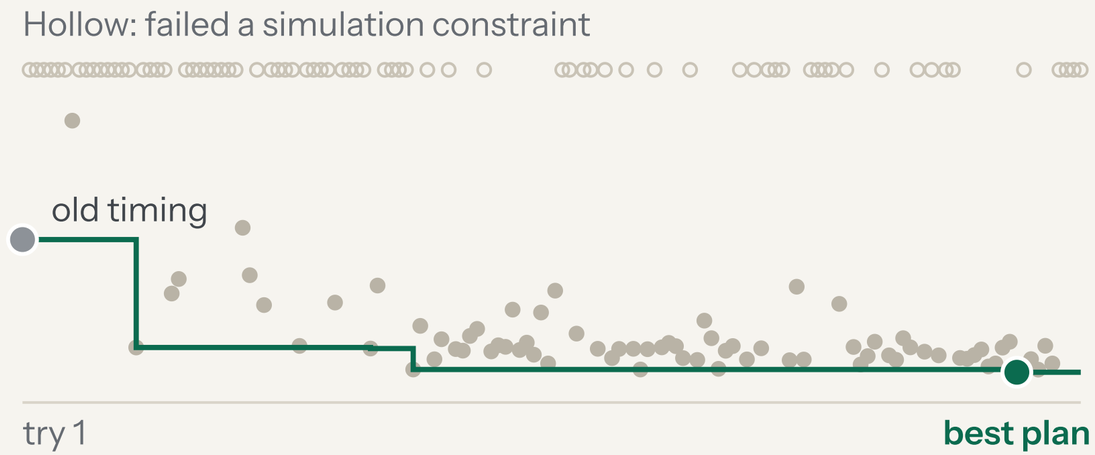
It learns the traffic pattern, then searches 150 timings on a practice copy.

1. LEARN: CARS PER 15 MINUTES, WOLFE RD



Learned from 3 watched rush hours. Darker means busier.

2. SEARCH: 150 TRIES ON THE PRACTICE COPY



Each dot is one try. Lower is better.

Model-based search (Optuna). After each try, it updates its guess of which settings look good and tries there next.

Why this approach? A bounded search leaves a record of each candidate and the rule that accepts or rejects it.

SIMULATED

One card: old and new settings, the reason, and the expected effect.

SIGNAL	CYCLE, S	OFFSET, S	MAIN GREEN ON WOLFE RD, S	REASON
Vallco Pkwy	65 → **88**	30 → **50**	22 → **38**	Learned peak: 408 cars per 15 min
I-280 south ramps	65 → **88**	0 → **58**	31 → **42**	Learned peak: 360 cars per 15 min
I-280 north ramps	65 → **88**	20 → **4**	21 → **35**	Learned peak: 502 cars per 15 min
Pruneridge Ave	65 → **88**	36 → **83**	10 → **20**	Learned peak: 344 cars per 15 min
Apple Park Way	65 → **88**	42 → **47**	17 → **34**	Learned peak: 360 cars per 15 min

Practice-copy forecast: 137.6 to 79.6 s (−42%). Offsets change with the longer cycle.

Unchanged: the order of the lights, yellow and all-red times, and minimum greens.

ENGINEER DECISION

☐ Approve ☐ Reject

HOW WE KEPT THE TEST FAIR

The learner never saw the road's true traffic, and the rules came first.

WHAT THE LEARNER GETS

- The road map
- The list of signals
- The old timing
- The sensor feed: car counts every 5 minutes

HIDDEN FROM THE LEARNER

The true traffic: every trip and every route.

A test fails if learner code ever opens it.

Rules written first. Frozen in our code history before the test ran.

Two roads held out until the test. No learner run before the test, one try each.

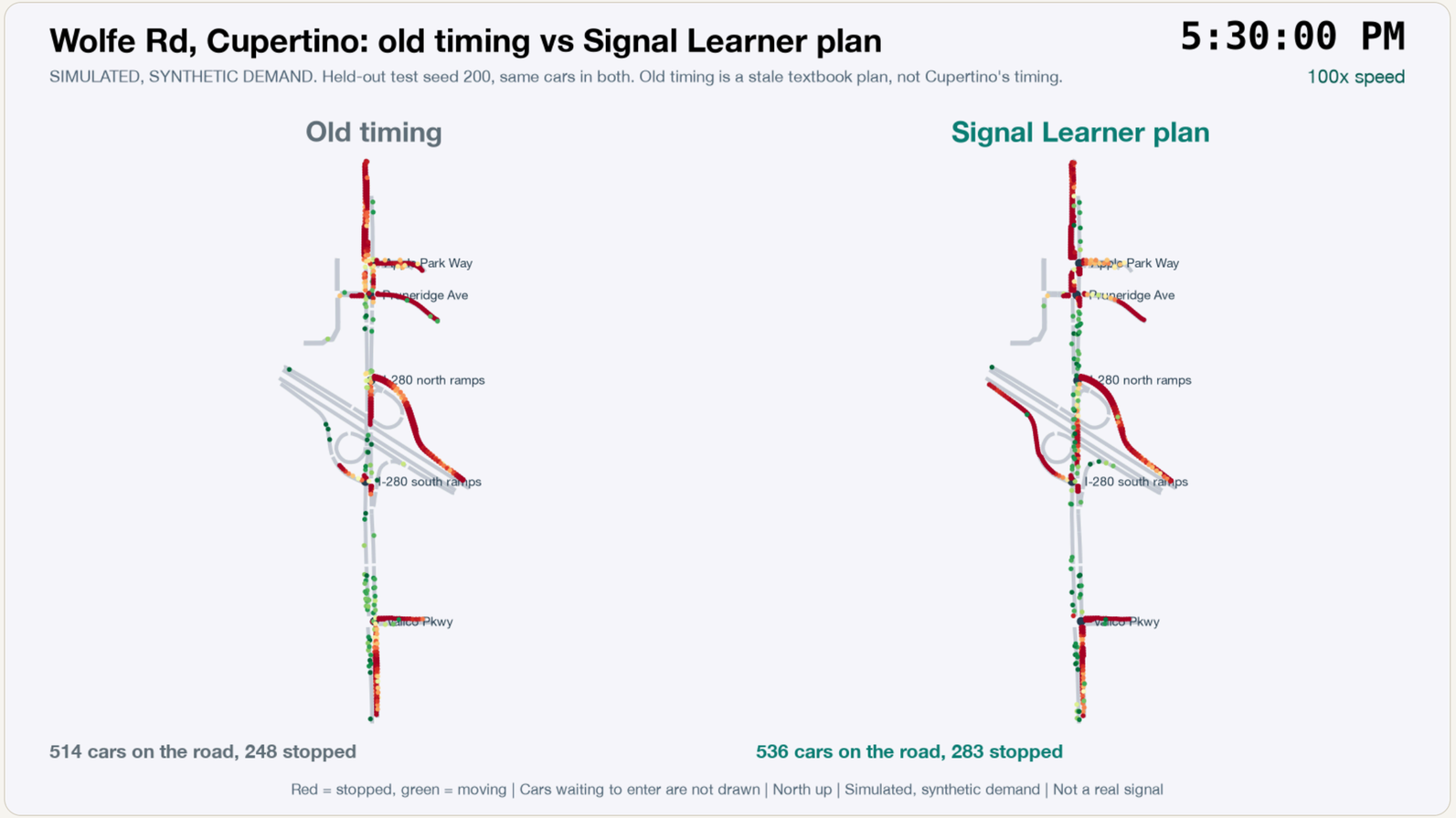
Paired runs. Each plan faced the same 20 held-out demand seeds.

Saved evidence. The original results stay frozen for review.

WATCH IT: WOLFE RD

SIMULATED

Same cars, same rush hour: old timing on the left, the learner's plan on the right.



SIMULATED, SYNTHETIC DEMAND

One test rush hour, played 100 times faster than real life.

Red dots are stopped cars. Green dots are moving.

Why the right side can show more cars: cars still waiting to get onto the road are not drawn. Old timing leaves more of them stuck outside.

RESULT 1: WOLFE RD, A HELD-OUT ROAD

SIMULATED

On held-out Wolfe, the historical plan lowered the delay score by 26.1%.

Old timing



870.49 s

Textbook retiming (perfect counts, no engineer tuning)



846 s −3%

Signal Learner (saw only the sensors)



643.11 s −26.1%

Delay score per requested trip: total completed-trip delay plus 900 s per unfinished trip and stuck-car removal, divided by all requested trips.

TRIPS THAT NEVER FINISHED, PER RUSH HOUR

968 to 217

Better on all 20 test rush hours.

WHAT WAS COMPARED

20 paired synthetic runs. Both baselines are constructed plans; neither is the city's actual timing.

RESULT 2: DE ANZA BLVD, THE SECOND HELD-OUT ROAD

● SIMULATED · POST-HOC CHECK

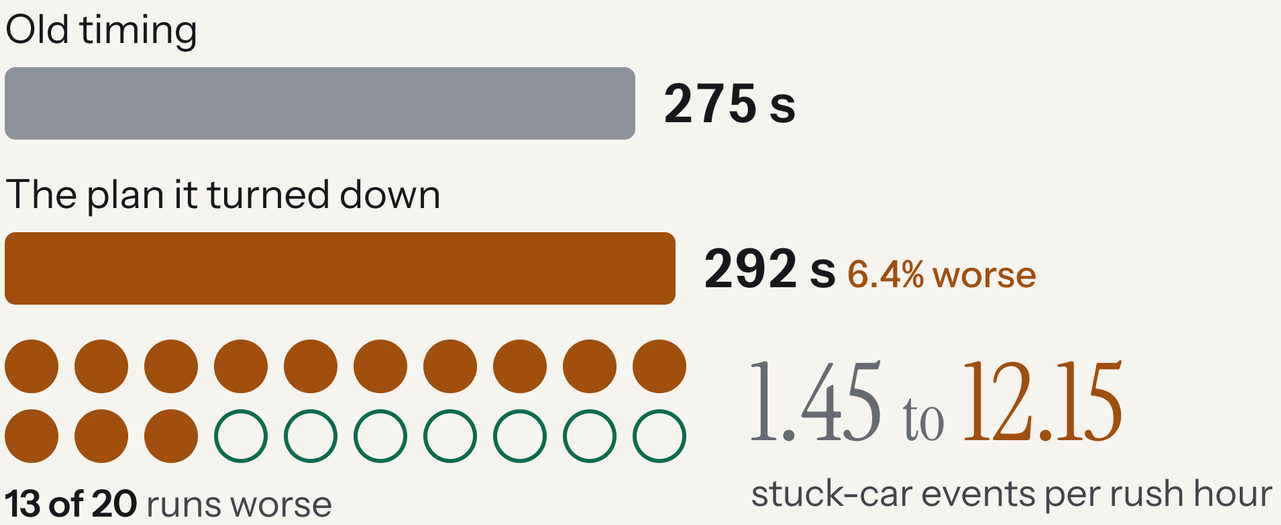
It said “no change”; a later check found the rejected plan was worse.

THE CARD SAID

No change.

Old timing stays. Knowing when to do nothing matters.

POST-HOC: CHECKED AFTERWARD ON 20 NEW RUNS



The practice copy was badly wrong here. One lane is shared by straight and right-turning cars, and the copy estimated 3,791 right-turn passages. It expected 1,740 s of delay; the simulated world had about 270 s. A teleport constraint rejected the candidate in the historical copy.

ALL FOUR ROADS

SIMULATED

Two development roads. Two held-out roads. Keep those results separate.

ROAD	USED FOR	TRAFFIC	OLD TIMING	TEXTBOOK RETIMING, PERFECT COUNTS	SIGNAL LEARNER	V5, SAW THE TRUE TRAFFIC
Homestead Rd	PRACTICE	Synthetic	130 s	113 s −13%	**77 s −41%**	not run
Stevens Creek Blvd	PRACTICE	VTA 2024 counts	682 s	661 s −3%	**471 s −31%**	**369 s** −46%
Wolfe Rd	HELD-OUT TEST	Synthetic	870 s	846 s −3%	**643 s −26%**	not run
De Anza Blvd	HELD-OUT TEST	Synthetic	270 s	481 s +78%	270 s no change	not run

Delay score per requested trip, lower is better. Our rules were tuned on the practice roads (10 simulated rush hours each), so only the held-out roads (20 each) are a fair test.

v5 is our earlier optimizer. It had full access to the true traffic and still wins on Stevens Creek. Stevens Creek uses VTA 2024 counts. **Source: VTA, 2024 CMP Monitoring and Conformance Report.**

LIMITS AND TRADE-OFFS

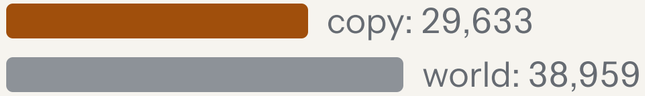
A better average is only the start of validation.

1 **Simulation, not street benefit.**
No live inputs or signal control. Held-out demand is synthetic; timing baselines are constructed.

2 **The worst origin metric rose.**
Wolfe: about 2,566 to 2,769 s, +203 s. Overall improvement hid a worse-served group.

3 **Shared lanes hide turn choices.**
De Anza's copy guessed too many right turns. Counts alone may not identify the split.

4 **The copy missed signal passages.**
Wolfe: 29,633 vs 38,959, about 24% fewer signal-turn passages, not unique vehicles.



5 **Rollback had false alarms.**
De Anza undid a harmless change in 4 of 10 runs. Normal variability needs better testing.

6 **Two held-out roads are a small test.**
New screening needs separate validation. Engineer comparison and real data remain open.

HOW THIS CONNECTS TO CUPERTINO

A proposed path to city use, with no integration built yet.

STEP 1

Read

Ask an engineer which count exports, timing files and logs can be shared. Validate an adapter on those files.

STEP 2

Recommend

Compare recommendations with an engineer's plan. Let the engineer assess constraints and enter any approved change.

STEP 3, MUCH LATER

Automatic

Controller integration would need city approval, a real interface, safety review and field validation. It is outside this prototype.

WHERE WE ARE

City email, Nov 1: Stevens Creek uses time-of-day timing plans.

No city timing files or live feed have been tested.

FRESH VALIDATION STATUS

Fresh checks exposed limits; original results unchanged.



Bubb could not run

Preparation failed because boundary routing is unsupported. This is a capability gap, not a successful test.



Stelling abstained

Occupancy mismatch and shared-turn ambiguity blocked search. Model-based (TPE) and random search both ran zero optimization trials.



Rollback missed harm

Fresh controls missed harmful changes. The rule remains experimental, not a safety mechanism.

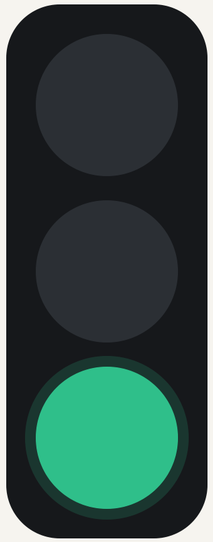


Use real city data

With permission, validate representative days and compare against actual city timing with an engineer.

In simulation, Signal Learner found a lower-delay plan on one held-out road and kept old timing on another.

Help an engineer *test the next step.*



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